

Demand Forecasting and Revenue Optimization for an E-Commerce Marketplace

Final Report for the BDM Capstone Project

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1 Executive Summary

This report analyses Olist, a Brazilian B2C e-commerce marketplace connecting independent third-party sellers to consumers nationwide, which processed 99,441 orders between September 2016 and October 2018. As a decentralised platform with thousands of independent sellers and no in-house logistics fleet, Olist faces three pressing problems: unpredictable month-to-month demand, delivery delays strongly linked to falling satisfaction, and no structured segmentation across its 96,096 unique buyers.

Nine interrelated CSV files covering payment_value, freight_value, review_score, order_status, and delivery dates were cleaned and analysed. Key statistics show a mean payment_value of BRL 160.99 (median BRL 105.29), a mean review_score of 4.09/5, and a 6.8% late-delivery rate. Four methods were applied in sequence: exploratory data analysis, time-series trend modelling, RFM-based K-Means clustering (k=4), and multiple linear regression for delivery-delay prediction.

The time-series analysis confirmed a predictable November demand spike. RFM clustering produced four segments: High-Value Champions (n=2,418, avg BRL 1,161), Loyal Customers (n=2,772), New/Recent Customers (n=50,642), and Dormant Customers (n=37,526). The regression model ($R^2=0.026$) identified Seller: São Paulo (+2.02 days) as the strongest delay predictor, while review scores fall sharply from 4.03 to 2.99 once delivery slips from on-time to slightly late.

The report recommends a 14-day re-engagement campaign for New/Recent Customers, a VIP programme for High-Value Champions, and delivery-window recalibration with logistics-hub decongestion. If implemented, these measures are projected to lift the late-delivery rate toward 4.0–4.5% and the average review score toward 4.20–4.30, while converting even a small share of the 88,168 one-time buyers into repeat customers would materially improve retention-driven revenue.

2 Proof of Originality

This study uses publicly available secondary datasets obtained from Kaggle. All datasets were accessed and downloaded independently for this academic project.

Field	Details
Dataset Title	Olist Brazilian E-Commerce Public Dataset

Primary Dataset URL	https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce
Supplementary Dataset	https://www.kaggle.com/datasets/olistbr/marketing-funnel-by-olist
Dataset Creator	Olist (https://olist.com)
Date Accessed	April 2025
Records	99,441 orders across 9 CSV files, September 2016 to October 2018
Analysis Notebook	https://github.com/23f3000862-svg/bdm-capstone-olist

All data-cleaning decisions, variable selections, visualisations, model analyses, findings, and recommendations were developed independently. No output has been copied from Kaggle notebooks or reports prepared by other individuals. The linked GitHub repository contains the Python notebook reproducing every figure and table in this report, including all charts referenced in Section 5.

3 Metadata and Descriptive Statistics

3.1 Dataset Metadata

The Olist dataset comprises nine interrelated CSV files documenting the full e-commerce transaction lifecycle. Table 1 provides a variable dictionary for the key variables used in this analysis, including meaning, data type, unit, value range, and business relevance. All charts and visualisations referenced in this section are presented together in Section 5 (Results and Findings), numbered sequentially as Figures 1 through 12.

Table 1: Variable Dictionary for the Olist E-Commerce Dataset

Variable Name	Meaning	Data Type	Unit	Range / Values	Business Relevance
payment_value	Total amount paid by customer for the order	Continuous	BRL	0 – 13,664	Monetary feature in RFM model; primary revenue variable
price (order)	Aggregate product price across all items in order	Continuous	BRL	0.85 – 13,440	Regression predictor for delivery delay
freight_value	Shipping cost charged to the customer per order	Continuous	BRL	0 – 1,795	Proxy for shipment weight; strongest regression predictor
review_score	Customer satisfaction rating submitted after delivery	Ordinal	Rating 1–5	1, 2, 3, 4, 5	Primary KPI for delivery performance analysis
delay_days	Days by which actual delivery exceeds estimated date	Continuous	Days	–147 to +188	Regression target variable; negative = early, positive = late

order_status	Current fulfilment stage of the order	Categorical	Category	delivered (97.0%), shipped, cancelled	Non-delivered orders excluded from delay analysis
payment_type	Payment method selected by the customer	Categorical	Category	Credit Card (73.9%), Boleto, Voucher, Debit	Payment composition; relevant to average order value
customer_state	Brazilian state where the customer is registered	Categorical	State code	27 states; SP=42.0%, RJ=12.9%	Geographic feature for same-state routing indicator
seller_state	Brazilian state where the seller is located	Categorical	State code	23 states; SP=59.7% of sellers	Seller: SP and Same-State regression predictors
item_count	Number of individual items included in the order	Discrete	Count	1 – 21 (mean 1.14)	Regression predictor; multi-item orders tend to be fulfilled faster
payment_installments	Number of monthly instalments chosen by customer	Discrete	Count	0 – 24 (mean 2.85)	Indicates consumer credit behaviour and purchase planning

3.2 Descriptive Statistics

Table 2 presents the distributional properties of all key numerical variables. The two formulae used throughout this report are defined as Equations 1 and 2.

$$\text{Late Delivery Rate} = (\text{Orders Delivered After Estimated Date} \div \text{Total Delivered Orders}) \times 100$$

Equation 1: Late Delivery Rate Formula

$$\text{Delivery Delay (days)} = \text{Actual Delivery Date} - \text{Estimated Delivery Date}$$

Equation 2: Delivery Delay Variable Definition

Applying Equation 1: $(6,564 \div 96,470) \times 100 = 6.8\%$ of delivered orders arrived late. Among late orders, the average overrun was 10.62 days. Overall, 93.2% of deliveries arrived ahead of schedule with an average early arrival of 13.51 days. Delay is computed at whole-calendar-day granularity (delivery date vs. estimated date); using sub-day timestamp precision instead shifts the late-delivery rate to approximately 8%, since it also counts deliveries that arrived a few hours past the estimated date as late. This report uses the calendar-day definition throughout, as the `estimated_delivery_date` field itself carries no time component.

Table 2: Descriptive Statistics for Key Numerical Variables

Variable	Unit	N	Mean	Median	Std Dev	Min / Max
payment_value	BRL	99,440	160.99	105.29	221.95	0 / 13,664
price (order level)	BRL	98,666	137.75	86.90	210.65	0.85 / 13,440
freight_value	BRL	98,666	22.82	17.17	21.65	0 / 1,795
review_score	1–5	99,224	4.09	5.00	1.35	1 / 5
delay_days	Days	96,470	−11.88	−12.00	10.18	−147 / +188
item_count	Count	98,666	1.14	1.00	0.54	1 / 21
payment_installments	Count	103,886	2.85	1.00	2.69	0 / 24

The `payment_value` mean (BRL 160.99) substantially exceeds the median (BRL 105.29), indicating a right-skewed distribution driven by a small number of high-value orders. The `review_score` distribution is left-skewed (mean 4.09, median 5.00, SD 1.35), reflecting that most customers are highly satisfied while a meaningful minority rate very low. The `delay_days` mean of −11.88 days confirms the platform systematically delivers ahead of schedule.

4 Detailed Explanation of Analysis Process and Method

4.1 Data Cleaning and Preprocessing

The raw dataset was examined for missing values, duplicates, and data-type inconsistencies before any analysis began. Han, Kamber, and Pei (2011) identify missing-value treatment and feature standardisation as the two most consequential preprocessing decisions in data-mining workflows [1]. Three timestamp columns in the orders table had structurally missing values corresponding to undelivered orders: `order_approved_at` (160 missing, 0.16%), `order_delivered_carrier_date` (1,783, 1.79%), and `order_delivered_customer_date` (2,965, 2.98%). These were not imputed but excluded from delivery analysis. No duplicate `order_id` entries were found. The payments table contained 103,886 rows for 99,441 orders because multiple payment instalments can exist per order; these were aggregated to a single per-order total using SUM. A regression working dataset of 96,470 valid records was constructed by joining the orders, items, customers, and sellers tables and removing rows with null feature values. For clustering, RFM features were computed per unique customer from the 96,478 delivered orders and z-score standardised prior to K-Means.

4.2 Analysis Workflow and Methods

The analysis followed ten structured steps: (1) Examine dataset structure and missing values. (2) Clean and preprocess as described above. (3) Compute descriptive statistics. (4) Conduct EDA across order volume, revenue, categories, payment types, and delivery performance. (5) Fit a monthly linear trend model to identify demand growth trajectory. (6) Engineer RFM features per customer. (7) Z-score standardise RFM features and apply K-Means clustering; select k via the Elbow Method. (8) Profile clusters and assign business labels. (9) Build a multiple linear regression model for delivery delay. (10) Derive recommendations from all findings.

Method 1 – Exploratory Data Analysis (EDA): Applied as the foundational first stage, following Tukey’s (1977) principle that systematic exploration of data structure must precede formal modelling [2]. Five areas were investigated: monthly order and revenue trends, day-of-week purchase patterns, product category revenue, payment method composition, and delivery timing vs. customer review score. This stage lays the groundwork for all three business objectives — demand patterns, delivery performance, and customer behaviour — identified in the Executive Summary.

Method 2 – Time-Series Linear Trend Model (addresses the demand-unpredictability objective): Monthly order volume was fitted with a linear trend model to quantify the platform’s growth rate. This method was selected over a moving-average filter because the goal is to describe trend slope, not smooth noise:

$$\hat{Y}(t) = \beta_0 + \beta_1 t$$

Equation 3: Linear Trend Model, where $\hat{Y}(t)$ is predicted order count in month t , β_0 is the intercept, and β_1 is the monthly growth coefficient.

Method 3 – RFM K-Means Clustering (addresses the customer-segmentation objective): Applied to segment customers using Recency, Frequency, and Monetary features. Following Kotler and Keller (2016), RFM is the standard framework for commercially significant retail segmentation [3]. All three features were z-score standardised before clustering (Equation 4) to prevent larger-scale variables from dominating Euclidean distances. K-Means minimises the Within-Cluster Sum of Squares (WCSS, Equation 5). K-Means was selected over hierarchical clustering because the 93,358-customer dataset makes hierarchical methods computationally infeasible:

$$z = (x - \mu) / \sigma$$

Equation 4: Z-Score Standardisation, where x is the observed value, μ is the variable mean, and σ is the standard deviation.

$$WCSS = \sum_i \sum_{\{x \in C_i\}} ||x - \mu_i||^2$$

Equation 5: WCSS Formula, where C_i is cluster i , μ_i is its centroid, and $||x - \mu_i||^2$ is squared Euclidean distance.

Method 4 – Multiple Linear Regression (addresses the delivery-delay objective): Constructed to quantify the marginal effect of operational and geographic variables on delivery delay. Following Wooldridge (2012), regression is the standard method for estimating marginal effects while controlling for confounders [4]. Regression was preferred over classification because the outcome is continuous and coefficient signs provide operationally interpretable insights:

$$\text{Delay} = \beta_0 + \beta_1(\text{Freight}) + \beta_2(\text{Price}) + \beta_3(\text{Items}) + \beta_4(\text{Month}) + \beta_5(\text{DOW}) + \beta_6(\text{CustSP}) + \beta_7(\text{CustRJ}) + \beta_8(\text{SellerSP}) + \beta_9(\text{SameState}) + \varepsilon$$

Equation 6: Multiple Linear Regression Model for Delivery Delay Prediction

5 Results and Findings

Analysis Notebook: All twelve figures below are generated directly from the Python analysis notebook, which is fully reproducible. Notebook: <https://github.com/23f3000862-svg/bdm-capstone-olist>

5.1 EDA Results (Figures 1–6)

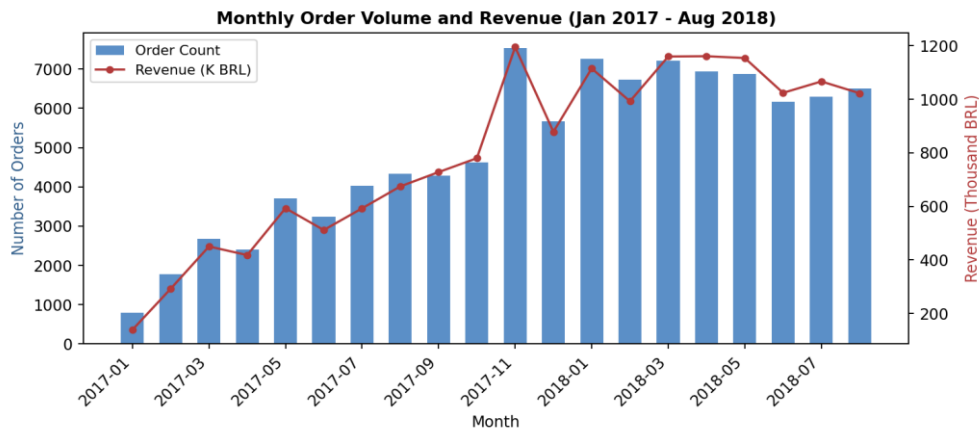


Figure 1: Monthly Order Volume and Revenue (January 2017 – August 2018)

Figure 1 uses a dual-axis combination chart to allow two variables on different scales to be compared on a shared time axis. A clear upward trend is visible, driven by a sustained rise in monthly order volume through 2017 into mid-2018. The November 2017 spike corresponds to Brazil’s Black Friday. A bar-and-line combination was selected because bars effectively show discrete monthly volumes while the line communicates the revenue trend across time.

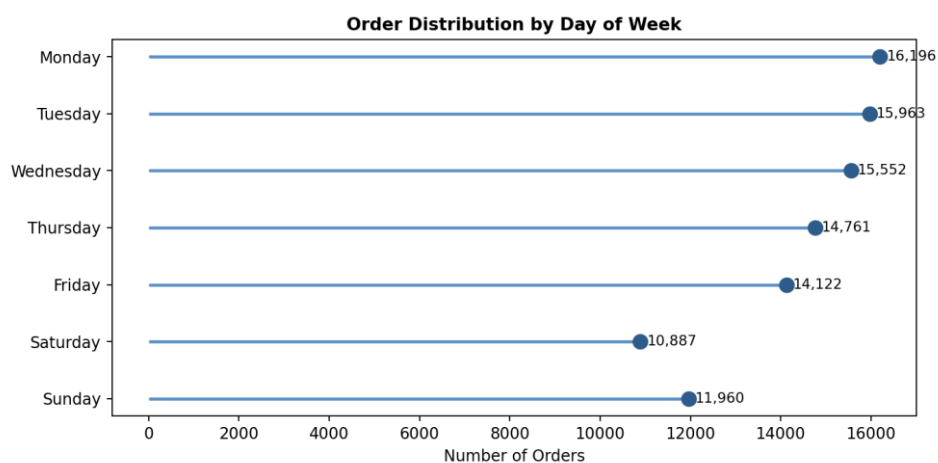


Figure 2: Order Distribution by Day of Week (Horizontal Lollipop Chart)

Figure 2 uses a horizontal lollipop chart because it encodes ranked categorical values more efficiently than a bar chart, reducing ink-to-data ratio while preserving precise value reading

through the dot position. Monday leads at 16,196 orders, roughly 35% higher than Sunday, with a clear mid-week decline pattern.

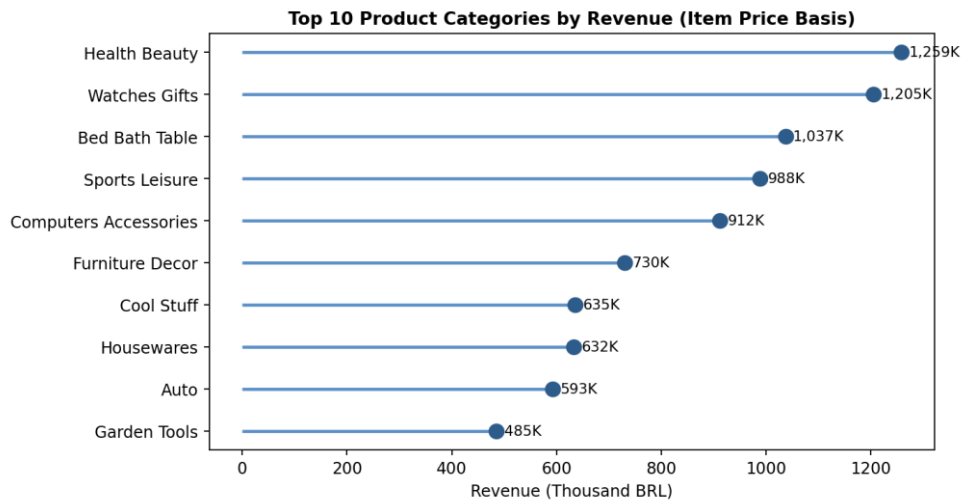


Figure 3: Top 10 Product Categories by Revenue (Lollipop Chart, Thousand BRL)

Figure 3 uses a lollipop chart for the same reason as Figure 2: ranked values across labelled categories are best shown with a format that does not require a bar to extend from zero when relative differences are the primary message. Health & Beauty leads at roughly BRL 1.26M. The top ten categories account for approximately 63% of total platform revenue.

Distribution of Payment Methods

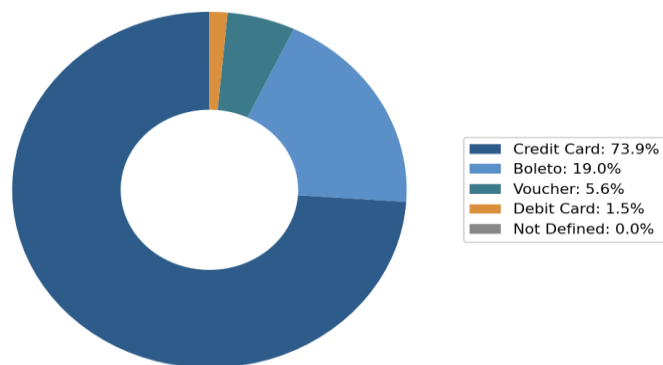


Figure 4: Distribution of Payment Methods

Figure 4 uses a pie chart because payment_type is a composition variable — the question is what share each method represents of the total, not a ranked comparison. Credit Card dominates at 73.9%, followed by Boleto (19.0%), Voucher (5.6%), and Debit Card (1.5%).

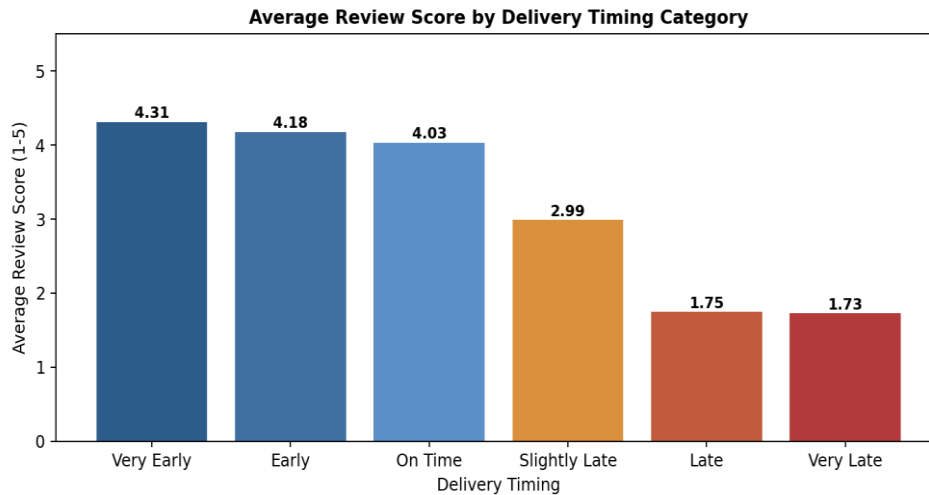


Figure 5: Average Review Score by Delivery Timing Category

Figure 5 uses a bar chart to facilitate direct comparison across six ordered timing categories. The discrete, named categories and a common zero baseline make bar format the most appropriate choice. A near-monotonic decline in review score is evident: very-early deliveries average 4.31, on-time arrivals 4.03, slightly-late orders 2.99, and very-late orders 1.73. This establishes delivery performance as the primary driver of customer satisfaction. Of 99,441 total orders, 96,478 (97.0%) were delivered; 93.2% arrived ahead of schedule.

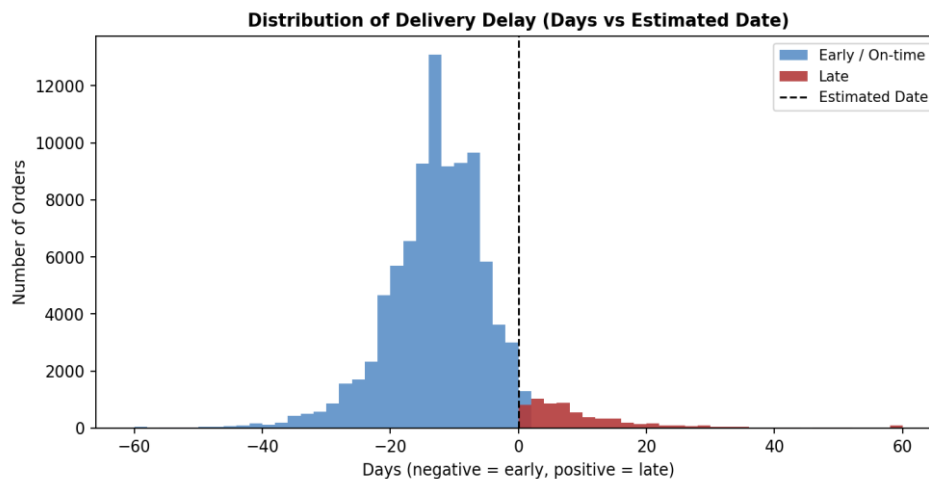


Figure 6: Distribution of Delivery Delay (Days vs Estimated Delivery Date)

Figure 6 uses a dual-colour histogram because the variable is continuous and the analytical question concerns the shape of two distinct sub-populations: early (blue) and late (red) deliveries. The distribution is strongly left-skewed; the right tail representing the 6.8% of late orders motivates the timing categories defined in Figure 5.

5.2 Time-Series Results (Figure 7)

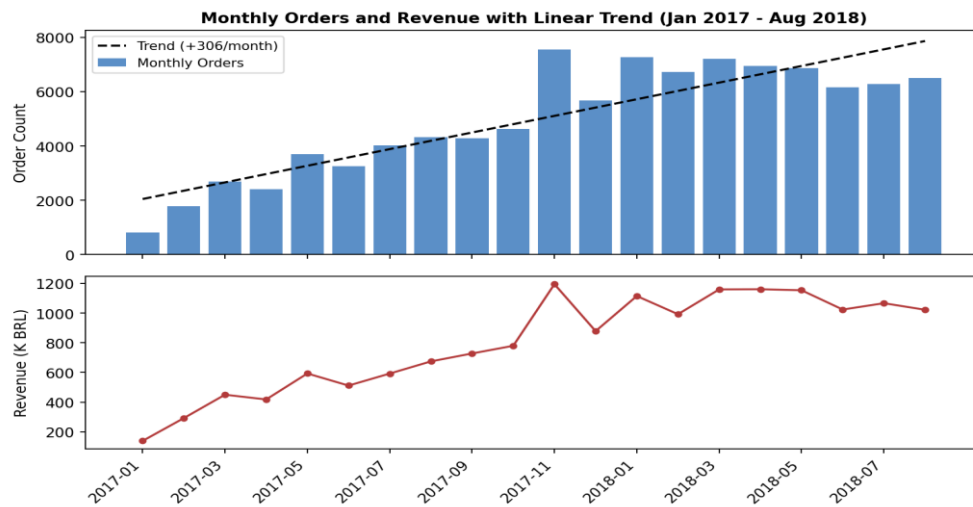


Figure 7: Monthly Orders with Linear Trend (January 2017 – August 2018)

Figure 7 overlays a fitted linear trend line on the monthly order-volume series to quantify platform growth. Two patterns are confirmed: (1) a sustained upward secular trend of approximately 307 additional orders per month; (2) a concentrated November peak driven by Black Friday, followed by a December–January dip reflecting post-peak demand normalisation. This predictable seasonal pattern means the platform can pre-position logistics capacity and issue seller readiness guidance roughly eight weeks ahead of the November surge.

5.3 Clustering Results (Figures 8–10)

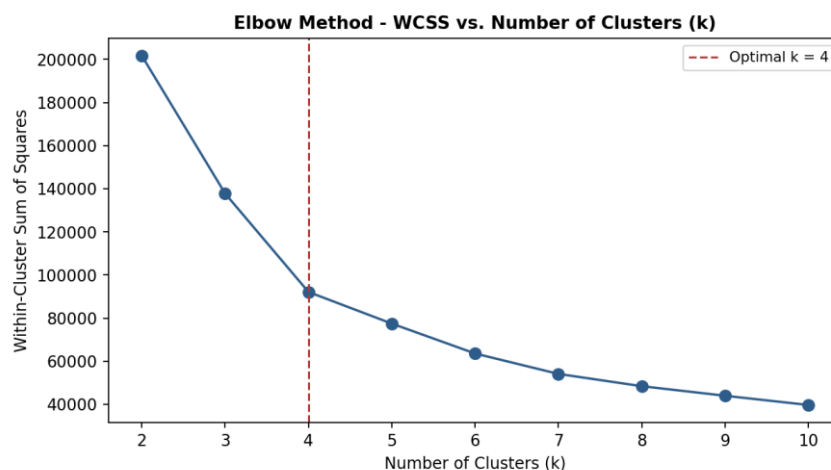


Figure 8: Elbow Method – Within-Cluster Sum of Squares vs. Number of Clusters (k)

Figure 8 uses a line chart because the interest is in the rate of change in WCSS across successive k values rather than a discrete comparison. A clear inflection (elbow) around $k=4$ confirms that

four clusters provide a good balance of within-cluster homogeneity and between-cluster separation.

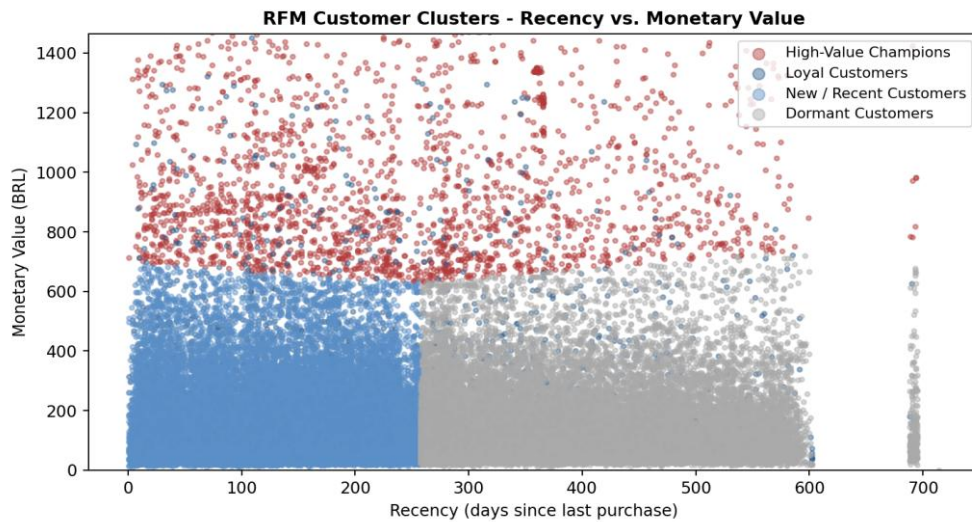


Figure 9: RFM Customer Clusters – Recency vs. Monetary Value (Scatter Plot)

Figure 9 uses a scatter plot because it allows all 93,358 individual customers to be visualised simultaneously, with two RFM dimensions encoded in position and cluster membership in colour. Four distinct groupings are apparent. The high concentration of light-blue points (New/Recent Customers) in the low-recency, low-monetary quadrant confirms this is the platform's largest segment.

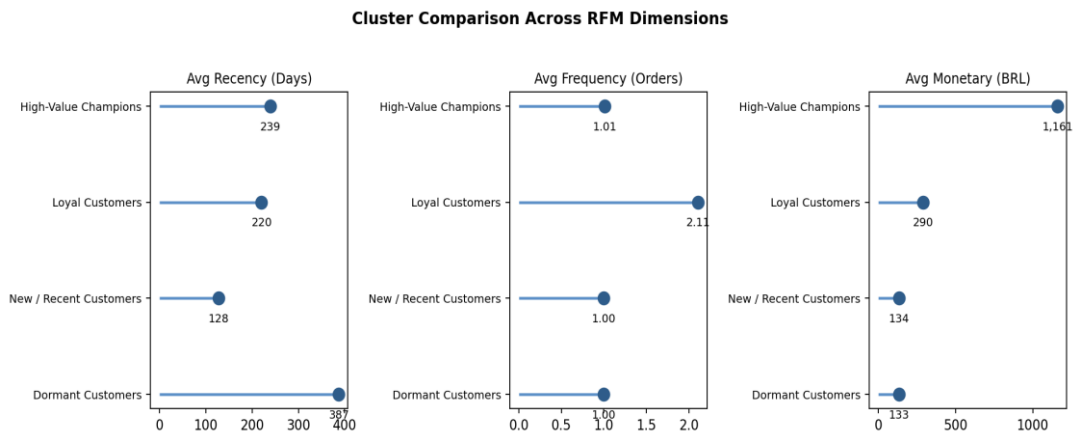


Figure 10: Cluster Comparison Across RFM Dimensions (Dot Plot)

Figure 10 uses a horizontal dot plot for each RFM dimension because dot plots encode a ranked comparison of continuous values more cleanly than bar charts by removing the visual weight of filled bars. The four cluster profiles are summarised in Table 3.

Table 3: RFM Cluster Profiles and Business Labels

Cluster	Business Label	n	Avg Recency (Days)	Avg Frequency	Avg Monetary (BRL)
Cluster 0	Dormant Customers	37,526	387	1.00	133
Cluster 1	Loyal Customers	2,772	220	2.11	290
Cluster 2	High-Value Champions	2,418	239	1.01	1,161
Cluster 3	New / Recent Customers	50,642	128	1.00	134

Cluster 0 (Dormant, n=37,526): Avg recency 387 days, frequency 1.0, spend BRL 133. Purchased once, over a year ago — primary win-back opportunity.

Cluster 1 (Loyal, n=2,772): Avg frequency 2.11, recency 220 days, spend BRL 290. The platform's most engaged repeat-buyer segment.

Cluster 2 (High-Value Champions, n=2,418): Avg monetary BRL 1,161 — roughly 7.0× the average spend across all clustered customers (BRL 165) and 7.2× the platform-wide mean order value (BRL 160.99). Highest-priority retention target.

Cluster 3 (New/Recent, n=50,642): Avg recency 128 days (most recently active), frequency 1.0. Largest segment and primary conversion opportunity.

5.4 Regression Results (Figures 11–12)

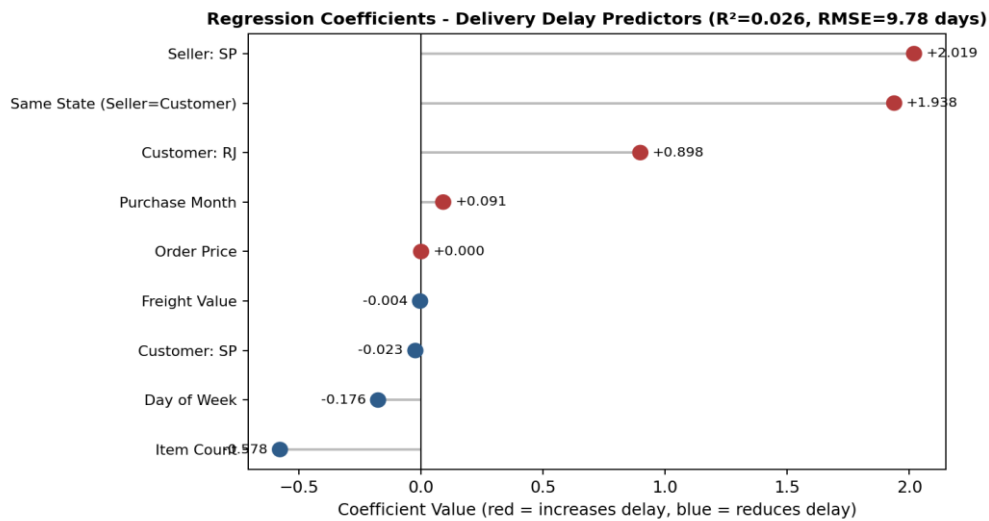


Figure 11: Regression Coefficients – Delivery Delay Predictors (Lollipop Chart)

Figure 11 uses a lollipop chart because the key information is the magnitude and direction of each coefficient rather than a comparison of filled areas. Red markers indicate variables that increase delay; blue markers reduce it. The model achieved $R^2=0.026$, $RMSE=9.78$ days. Both Seller: São Paulo (+2.02 days) and Same State (+1.93 days) carry positive coefficients. This counterintuitive result reflects logistics hub congestion: São Paulo handles the highest order concentration in Brazil, and congestion at the main distribution hub overrides any geographic proximity benefit. Item Count is negative (−0.58), indicating multi-item orders tend to be more strategically planned and fulfilled faster.

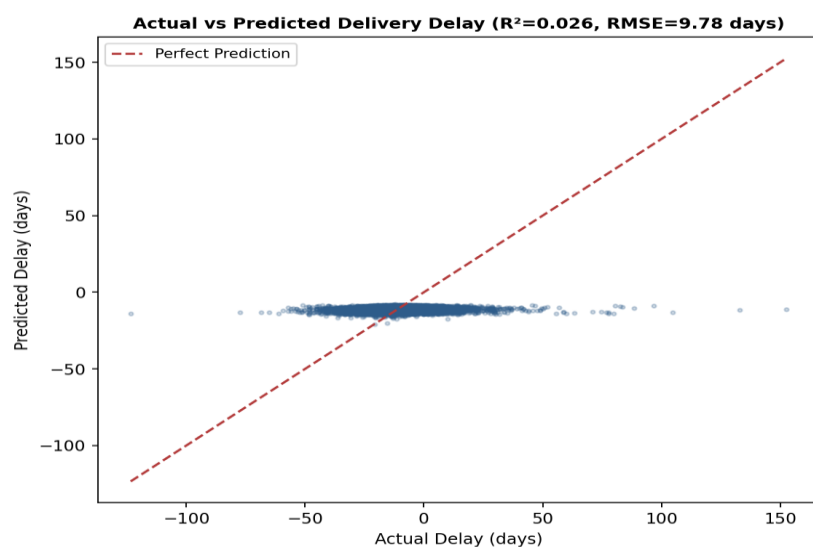


Figure 12: Actual vs. Predicted Delivery Delay (Scatter Plot, Hold-Out Test Set)

Figure 12 uses a scatter plot to show the joint distribution of actual and predicted delay values on the held-out test set. The red dashed perfect-prediction line enables immediate assessment of model calibration. The model captures central tendency well but underestimates extreme late-delivery cases, as expected from a linear model applied to a right-skewed outcome. This modest R^2 motivates the limitation discussed in Section 6.4: carrier-level routing data, not present in this secondary dataset, would likely explain most of the remaining variance.

6 Interpretation of Results and Recommendations

6.1 Overall Strategic Picture

Three findings define the strategic picture. First, delivery performance is the single most powerful determinant of satisfaction: the transition from on-time to slightly-late delivery (1–5 days) reduces average review score from 4.03 to 2.99, a decline no product or pricing variable can match (Figure 5). Reducing the 6.8% late-delivery rate is the highest-impact intervention available. Second, of the 93,358 customers with a valid RFM profile, 94.4% (Clusters 0 and 3, combined 88,168) are one-time buyers with average frequency 1.0. Converting them to repeat buyers is the primary growth lever. Third, the 2,418 High-Value Champions averaging BRL 1,161 per order are financially critical; a 10% churn rate in this segment equates to a material annual revenue loss.

6.2 Cluster-Wise Recommendations

1. **Cluster 3 – New / Recent Customers (n=50,642):** Deploy an automated re-engagement email within 14 days of first delivery featuring category recommendations based on the initial purchase, combined with a time-limited voucher on a second order. Target credit-card users (73.9% of transactions) to minimise cost. Goal: convert first-time buyers into Loyal Customers.
2. **Cluster 2 – High-Value Champions (n=2,418):** Introduce a VIP programme: priority fulfilment routing, proactive order-status notifications, and exclusive early access to seasonal promotions. Monitor recency and trigger an immediate personalised win-back campaign at the first sign of disengagement.
3. **Cluster 1 – Loyal Customers (n=2,772):** Target with cross-category promotions pairing their historical category with top-revenue categories (Health & Beauty, Watches & Gifts). Incorporate loyalty recognition messaging to reinforce repeat behaviour.
4. **Cluster 0 – Dormant Customers (n=37,526):** Launch a win-back campaign with a discount incentive and a pre-filled cart based on the previous purchase category to recover a meaningful share of this segment at low incremental cost.

6.3 Operational Recommendations

Logistics congestion management: The positive coefficients on Seller: São Paulo (+2.02 days) and Same State (+1.93 days) indicate hub congestion is a structural driver of delay. The

platform should work with primary freight carriers to identify route bottlenecks and introduce alternative routing options for high-volume São Paulo-origin orders.

Black Friday capacity planning: The November demand spike is predictable (Figure 7). The platform should issue seller capacity guidance at least eight weeks before November, communicating the expected volume uplift above the October baseline and recommending inventory build-up targets.

Delivery window calibration: The platform consistently over-estimates delivery windows (average early arrival 13.51 days). Mollenkopf, Frankel, and Russo (2011) demonstrated that accurate expectation-setting reduces dissatisfaction even when physical delivery time is unchanged [5]. Narrowing estimated windows for standard orders would shift a meaningful proportion of currently "on-time" deliveries to "early", lifting review scores without any operational change.

6.4 Limitations and Business Impact

- The dataset is secondary data without carrier-level routing information, which would substantially improve the regression R^2 (currently 0.026). Unobserved heterogeneity in carrier networks likely accounts for most of the unexplained delivery-delay variance.
- The study window covers only September 2016 to October 2018 and does not reflect post-pandemic Brazilian e-commerce dynamics.
- Product weight (product_weight_g) was available in the metadata but could not be reliably joined at the order level for the regression model; inclusion in future analysis would likely improve delivery-delay prediction.
- The high one-time-buyer rate may partly reflect platform or dataset-construction characteristics. The customer_unique_id field was used for deduplication to mitigate this.

Implementing the cluster-based re-engagement programme for Clusters 0 and 3 (88,168 customers) even at conservative single-digit conversion rates would generate a meaningful number of additional repeat orders. Addressing logistics congestion and delivery-window calibration is estimated to reduce the late-delivery rate from 6.8% toward 4.0–4.5%, moving the average platform review score from 4.09 toward the 4.20–4.30 range.

This project demonstrates that EDA, time-series analysis, RFM clustering, and regression, applied sequentially to a rich transactional dataset, produce a coherent evidence base. Delivery

timing is the dominant driver of customer satisfaction; one-time buyers represent the primary growth lever; and São Paulo logistics-hub congestion is a measurable and addressable predictor of delay. All code and notebooks reproducing Figures 1–12 are available at the GitHub repository cited in Section 2.

References

- [1] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. Waltham, MA: Morgan Kaufmann, 2011.
- [2] J. W. Tukey, *Exploratory Data Analysis*. Reading, MA: Addison-Wesley, 1977.
- [3] P. Kotler and K. L. Keller, *Marketing Management*, 15th ed. Upper Saddle River, NJ: Pearson, 2016.
- [4] J. M. Wooldridge, *Introductory Econometrics: A Modern Approach*, 5th ed. Mason, OH: Cengage Learning, 2012.
- [5] D. A. Mollenkopf, R. Frankel, and I. Russo, “Creating value through returns management,” *J. Operations Management*, vol. 29, no. 5, pp. 391–403, 2011.
- [6] Olist Brazilian E-Commerce Dataset – Kaggle. Available: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>
- [7] Olist Marketing Funnel Dataset – Kaggle. Available: <https://www.kaggle.com/datasets/olistbr/marketing-funnel-by-olist>
- [8] Scikit-learn – K-Means Clustering. Available: <https://scikit-learn.org/stable/modules/clustering.html#k-means>
- [9] D. Grewal, A. L. Roggeveen, and J. Nordfält, “The future of retailing,” *J. Retailing*, vol. 93, no. 1, pp. 1–6, 2017.
- [10] IBM – Business Problem to AI/Data Science Solution. Available: <https://www.ibm.com/garage/method/practices/discover/business-problem-to-ai-data-science-solution/>